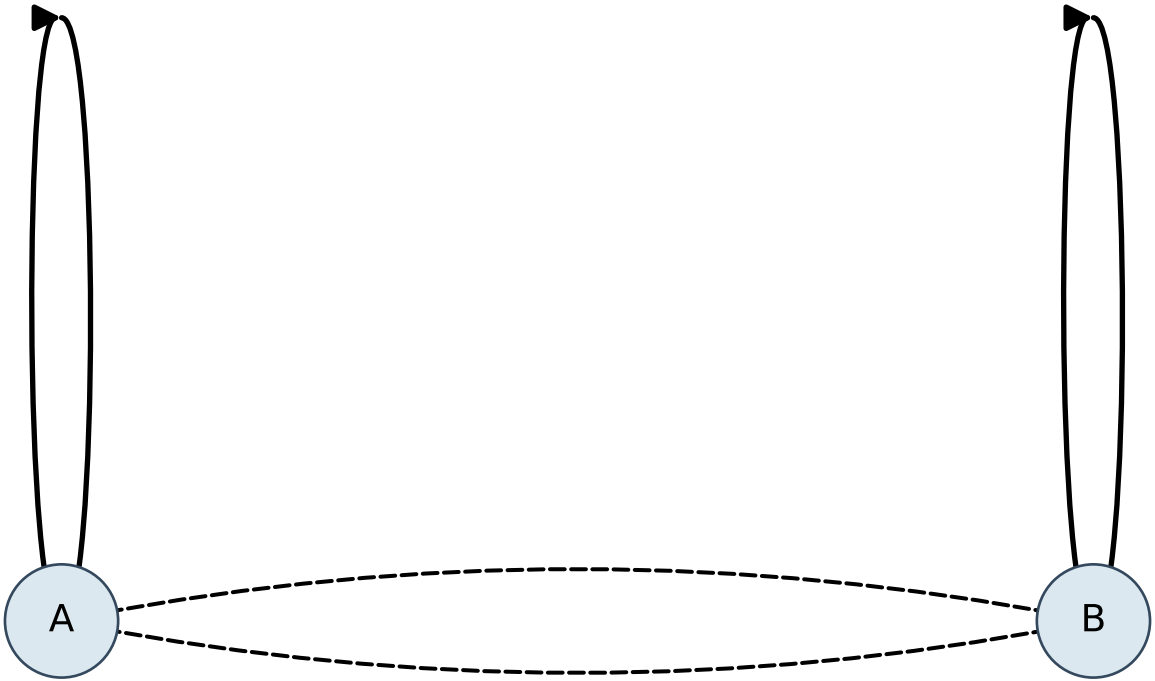Conceptual two-species catalytic core used in the toy model